Supporting Information

Definitional Instability in Kinase Inhibitor Selectivity: An Empirical Characterization and Motivated Properties for a Well-Formed Measure

Polina Vinogradova

This document reports, in full, four robustness checks summarized in the main text. All numbers are reproducible from the code repository (`https://github.com/polinavino/kinase-selectivity-definitions`); the script generating each table is named in its caption.

Table S1: Panel-size recoverability under alternative correlation bars

The main text calls a panel size "recovering" the full-panel ranking once mean Spearman ρ against the full panel exceeds 0.90, a convention borrowed from reliability research's "excellent agreement" bar rather than derived from the data (Methods, Results). Table S1 reports p^* (the smallest panel size reaching the bar) at four alternative bars, for both sparse-coverage datasets.

Table 1: Panel-size p^* (kinases) at four correlation bars. (`pstar_bar_sensitivity.py`)

Klaeger (343 kinases)	entropy	gini	s_score	ratio
bar = 0.80	80	260	50	320
bar = 0.85	80	260	80	320
bar = 0.90	110	290	110	343
bar = 0.95	140	320	200	343

Metz (172 kinases)	entropy	gini	s_score	ratio
bar = 0.80	50	50	50	170
bar = 0.85	50	50	50	170
bar = 0.90	80	80	50	170
bar = 0.95	110	110	80	170

The ratio remains the slowest to recover and the Gini coefficient the second-slowest for every bar tested; only the relative order of entropy and the S-score is bar-dependent, and only within $[0.80, 0.95]$.

Table S2: Bosc et al. window/ranking score and the CATDS proxy

Bosc et al.'s window score (WS) and ranking score (RS) and a CATDS-style proxy (Discussion) were computed on all four datasets to check whether they behave as additional members of the ratio family (WS, RS) or the distribution-based family (CATDS), as claimed in the Discussion. (`bosc_catds_comparison.py`)

Table 2: Spearman correlation of WS/RS against the ratio family, and of the CATDS proxy against the ratio, entropy, and Gini families.

Dataset	n_{cpd}	WS vs. ratio	RS vs. ratio	CATDS vs. ratio	CATDS vs. entropy / Gini
Davis	68	0.804	0.878	0.509	0.955 / 0.960
Klaeger	222	0.909	0.892	0.681	0.850 / 0.985
Anastassiadis	178	0.959	0.891	0.696	0.948 / 0.965
Metz	704	0.910	0.899	0.447	0.883 / 0.906

WS and RS correlate strongly with the ratio family on every dataset ($r = 0.80$–0.96), supporting their classification alongside $R(k)$. The CATDS proxy instead correlates more strongly with entropy and Gini than with the ratio on every dataset, supporting the Discussion's claim that CATDS is a distributional construction despite its historical classification as a ratio measure.

Table S3: Cross-definition correlations under the original metric operationalizations

Methods reports summary statistics for this check; the full 4×4 correlation matrices are given here. (`original_operationalization_robustness.py`)

On Klaeger and Metz, literal entropy correlates more strongly with the ratio than with either S-score or Gini – under this operationalization, entropy would not be classified in the distribution-based family at all on either sparse-coverage dataset. On Davis, literal entropy decouples from both families (consistent with Figure 1 of the main text). The native-Gini check on Anastassiadis is the one case that preserves the qualitative two-family pattern, albeit more weakly than the manuscript's baseline-swept form.

Table 3: Davis ($n = 68$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (complete matrix).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.417	0.952	0.456
entropy(literal)	0.417	1.000	0.424	0.336
Gini	0.952	0.424	1.000	0.343
ratio	0.456	0.336	0.343	1.000

Table 4: *
Klaeger ($n = 222$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (measured high-confidence entries only, per compound).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.570	0.722	0.223
entropy(literal)	0.570	1.000	0.688	0.789
Gini	0.722	0.688	1.000	0.616
ratio	0.223	0.789	0.616	1.000

Table 5: *
Metz ($n = 704$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (measured entries only, per compound).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.239	0.879	0.211
entropy(literal)	0.239	1.000	0.416	0.758
Gini	0.879	0.416	1.000	0.229
ratio	0.211	0.758	0.229	1.000

Table 6: *
Anastassiadis ($n = 178$): manuscript S-score/entropy/ratio vs. native %inhibition Gini (no baseline subtraction).

	S-score	entropy	Gini(native)	ratio
S-score	1.000	0.880	0.345	0.489
entropy	0.880	1.000	0.632	0.529
Gini(native)	0.345	0.632	1.000	0.314
ratio	0.489	0.529	0.314	1.000

Table S4: Candidate measure cross-dataset validation

Orientation (correlation with the number of active kinases, more negative is more correctly oriented toward "fewer active targets = more selective"), D3 robustness under the emphasis parameter β with the gate x_0 fixed (worst-case pairwise Spearman ρ), D3 robustness when x_0 itself is swept as a free parameter ("gate swept", the equivalent comparison to the entropy's baseline sensitivity; see Figure 6 of the main text), and the percentage of compounds for which D4 (monotonicity under a weak additional interaction) holds exactly. (`candidate_validation.py`)

Table 7: Candidate measure: orientation, D3 (two variants), and D4, across all four datasets.

Dataset	n_{cpd}	orientation r	D3, gate fixed	D3, gate swept	D4 (% compounds)
Davis	68	-0.969	0.998	-0.838	100.0
Klaeger	222	-0.936	0.999	-0.932	100.0
Anastassiadis	178	-0.936	0.997	$+0.913$	100.0
Metz	704	-0.791	0.991	-0.330	100.0

The "gate swept" column confirms, on three of four datasets, that the candidate's D3 satisfaction rests on fixing the gate x_0 by declaration rather than on the measure being inherently insensitive to its placement (worst-case ρ is far from 1, and on Klaeger and Davis is strongly negative). Anastassiadis is an exception ($\rho = +0.913$): on that dataset the candidate's ranking happens to be comparatively insensitive to the gate's placement as well, for reasons we do not have a mechanistic account of.

Smoothing-width T was also swept on Klaeger ($T \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$): orientation and D3-emphasis-robustness are essentially unchanged across this range ($r \approx -0.94$, $\rho \geq 0.989$), and the worst-case pairwise ranking agreement across all five values of T is $\rho = 0.999$. Removing the gate inverts the measure's orientation entirely (gated $r = -0.936$ vs. ungated $r = +0.943$ against n_{active} on Klaeger), which is why the gate, not merely the smooth emphasis, is required for D1 and correct orientation jointly.